Regional medullary redox profiles in rats with atlantoaxial joint disorder

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Summary

Atlantoaxial joint disorder has been associated with elevated blood pressure in a rat model, but its regional medullary biochemical phenotype remains incompletely characterized. We examined superoxide dismutase (SOD) activity and thiobarbituric acid (TBA)-derived malondialdehyde (MDA)-related signals, with *Nox1*/*Nox4*-targeted RT-qPCR as an exploratory measurement. Male Wistar rats underwent sham exposure or atlantoaxial implantation. Sham and model cohorts were sampled at week 4; a separate restoration cohort underwent implant removal at week 4 and sampling at week 8. Each endpoint included six independently processed rat samples per group. Tissue blocks contained the rostral ventrolateral medulla and adjacent medulla. At week 4, model rats had lower SOD activity than sham rats (0.564 ± 0.072 versus 1.122 ± 0.072 U/mg protein) and higher MDA-related signals (0.349 ± 0.034 versus 0.095 ± 0.017 nmol/mg protein; both adjusted $P < 0.001$). Normalized *Nox1*/*Nox4*-targeted assay estimates were higher in model samples, although unconfirmed product specificity precluded conclusions about transcript abundance. At week 8, restoration-cohort SOD and MDA-related values were 1.035 ± 0.101 and 0.213 ± 0.049, respectively. These later observations cannot isolate an implant-removal effect because sampling times differed and a matched second-surgery control was absent. The concurrent comparison identifies a regional medullary biochemical difference associated with this model. Assay-validation limitations and omission of postoperative analgesia constrain mechanistic interpretation.

Key words: Atlantoaxial joint; Medulla oblongata; Superoxide dismutase; Malondialdehyde; Oxidative stress

Introduction

An experimental model of atlantoaxial joint misalignment has been associated with increased arterial pressure in rats [1]. This model provides a means of investigating physiological changes accompanying an upper cervical joint disturbance. It does not, by itself, establish that cervical misalignment causes hypertension in humans. A related study examined angiotensin II (Ang II) and its type 1 receptor (AT1-R) in the rostral ventrolateral medulla (RVLM) of rats with this disorder [2]. Blood-pressure observations from the underlying experimental program were also used in that report. The present study addresses regional biochemical and *Nox1*/*Nox4*-targeted assay observations rather than presenting the shared blood-pressure measurements as new efficacy evidence.

The RVLM contains bulbospinal neurons involved in sympathetic vasomotor regulation [3]. Its contribution to blood-pressure control depends on the physiological state, including oxygen availability, baroreceptor input and anesthesia [4]. These observations make the ventrolateral medulla relevant to investigation of the cervical model, but they also caution against assuming that any regional biochemical difference directly represents increased sympathetic output. Anatomical proximity and physiological plausibility require experimental testing before they can be joined into a causal pathway.

Redox regulation is one candidate process. In spontaneously hypertensive rats, reduced RVLM antioxidant-enzyme activity has been linked experimentally to hypertension, and local antioxidant-enzyme gene transfer lowered arterial pressure [5]. Ang II-induced pressor responses have also been linked to NADPH oxidase-dependent signaling in the RVLM [6]. Mitochondrial processes provide another experimentally supported route to regional redox disturbance [7]. These findings concern other experimental settings and do not identify the source of an oxidative signal in the atlantoaxial model. NOX1 and NOX4 are nevertheless reasonable candidate molecules because their regulatory and subcellular properties differ [8].

A useful initial question is whether this model shows a detectable regional biochemical phenotype. We therefore examined SOD activity and TBA-derived MDA-related signals in tissue containing the RVLM and adjacent medulla. These measurements address antioxidant activity and an oxidation-related biochemical readout, respectively, while retaining the limitations of their detection chemistry [9]. *Nox1*/*Nox4*-targeted RT-qPCR was examined as an exploratory molecular measurement. The concurrent week-4 model–sham comparison is the principal comparison emphasized in this report. A separate cohort sampled at week 8 after implant removal provides a later observation, without establishing the specific effect of removal. No single primary endpoint was specified before the experiment.

Methods

Animals and experimental design

The parent experiment was approved by the Animal Ethics Committee of Anhui University of Chinese Medicine (AHUCM-rats-2022032). Male Wistar rats, approximately 12 weeks old and weighing approximately 250 g, were obtained from the Laboratory Animal Center of Anhui Medical University. Rats were housed individually at 22 ± 1°C and 60% relative humidity under a 12 h light/12 h dark cycle, with unrestricted food and water. Acclimatization lasted 1 week. Environmental enrichment was not documented. The use of males followed the earlier model study; sex-dependent effects were not assessed.

Sixty rats were initially allocated by random numbers to sham operation (n = 10) or model induction (n = 50). Six modeled rats died intraoperatively and three postoperatively. Of 41 survivors, seven lost their implants and 34 retained them. After screening, 30 successfully modeled rats were allocated by random numbers to model, restoration or superior cervical ganglion resection cohorts (10 each); four were retained in reserve. Only sham, model and restoration cohorts are considered here. Each reported biochemical and RT-qPCR endpoint used samples from six rats per group, selected from the corresponding cohort and processed separately without cross-animal pooling. The historical animal–sample ledger and cross-assay overlap could not be recovered; biological n and same-animal target/reference pairing were confirmed by the investigators.

The design was exploratory, without an outcome-based a priori sample-size calculation or a prespecified single primary endpoint. No retrospective power calculation was performed. Blinding was not documented. Sham and model rats underwent terminal sampling at week 4. Restoration rats were independently allocated from the successfully modeled population, underwent implant removal at week 4 and were sampled at week 8. Terminal tissue comparisons therefore involved different animals, and no week-8 time-matched model or second-surgery sham cohort was available.

Model induction and perioperative care

An S-shaped aluminum implant was inserted into the left atlantoaxial joint through an anterior cervical approach, following the previously described model [1]. Sham rats underwent exposure without implantation. Model success required a retained implant, asymmetric joint morphology with a left–right odontoid–lateral-mass distance difference >0.05 mm, and systolic pressure at least 20% above baseline with a plateau by week 4. Screening used digital radiography and tail-cuff pressure measurements. Restoration rats underwent re-exposure and implant removal at week 4.

Model induction, implant removal and terminal sampling used inhaled isoflurane at 2%, with a gas flow of 1 L/min. Anesthetic depth was monitored through respiration and reflex responses. Penicillin sodium was administered intramuscularly at 400,000–800,000 units/kg once daily for 3 days after surgery. No postoperative analgesics were administered, and no specific written justification or authorization for their omission was retained. Food intake, activity and wound condition were checked daily for the first 3 postoperative days and 2–3 times weekly thereafter. Potential contributors to the nine deaths included anesthesia-related complications, surgical injury and infection, but individual causes were not adjudicated in the retained records.

Tissue collection and biochemical measurements

After abdominal aortic sampling under isoflurane, rats received an overdose of inhaled isoflurane. Death was confirmed by the absence of spontaneous breathing, palpable cardiac activity and withdrawal to a noxious stimulus before brainstem removal. Atlas-guided bilateral regional blocks, approximately 100 mg per rat, contained the RVLM and adjacent medullary tissue. They were not isolated RVLM nuclei. Tissue was ground in liquid nitrogen and lysed on ice in RIPA buffer containing 1 mmol/L PMSF. Supernatants were collected after centrifugation, and protein concentration was measured by BCA assay (Beyotime, P0010).

Total SOD activity was measured by a WST-1 assay (Nanjing Jiancheng, A001-3). The reaction was incubated at 37°C for 20 min, and absorbance was read at 450 nm using an RT-6000 reader. Blank-corrected inhibition was converted using the reaction-to-sample volume ratio and the corresponding protein concentration, yielding U/mg protein. MDA-related signals were measured by the TBA assay (Nanjing Jiancheng, A003-1). Reaction mixtures were heated at 95°C for 40 min, cooled and centrifuged at 3500–4000 rpm for 10 min. Absorbance at 530 nm was converted using a 10 nmol/mL standard and normalized to protein concentration, yielding nmol/mg protein. The TBA result was interpreted according to its limited chemical specificity [9]. Detailed calculations and assay qualifications are provided in the Supplementary Information.

An O12 probe-associated fluorescence measurement was retained as an auxiliary observation in the Supplementary Information. The serum-designated assay had not been validated for the tissue/RIPA matrix. It was not used to determine an absolute ROS concentration or to identify a ROS species or source. Archived hematoxylin–eosin fields were also restricted to supplementary illustration because specimen counts and image-to-animal correspondence could not be confirmed. Neither observation was used to establish a quantitative histological or mechanistic conclusion.

Exploratory targeted RT qPCR

RNA was extracted using TRIzol, and reverse transcription used PrimeScript RT reagent Kit with gDNA Eraser (RR047A). SYBR reactions were run on a PikoReal 96 system. The program comprised 95°C for 1 min and 40 cycles of 95°C for 20 s and 60°C for 1 min, followed by melting analysis from 60°C to 95°C in 0.2°C increments. Primer sequences, expected product lengths and reaction composition are given in the Supplementary Information. Each target and β-actin reference was measured once per sample, without technical replication.

Within-sample ΔCq was the target Cq minus its paired β-actin Cq. Relative quantity was calculated as 2 raised to −ΔCq and divided by the arithmetic mean quantity of the six sham samples for that assay. This normalization assumes equal, approximately 100% amplification efficiencies, which were not experimentally validated. Reference-gene stability was not validated, and NTC/no-RT records were unavailable. Heterogeneous melt profiles and absent product-identity evidence left target specificity unconfirmed [10]. Two plate-annotation errors, B10-Ch5 and D12-Ch5, were explained by the investigators after review of loading/laboratory records; the original exports and numerical assignments were preserved. Details are in Supplementary Table S4. The results are reported as exploratory normalized assay estimates.

Statistical analysis

The rat was the experimental unit. Results are mean ± sample SD, with six samples per group per endpoint. All supplied observations were included without outlier removal or imputation. Biochemical comparisons used Welch one-way ANOVA and Games–Howell tests with 95% confidence intervals; adjustment covered three pairwise comparisons within each endpoint, without adjustment across endpoints. RT-qPCR inference used two-sided Welch t tests on ΔCq, with Holm correction across the full original family of 18 comparisons from six targets and three group contrasts. The correction family was retained although only *Nox1* and *Nox4* are displayed. $P < 0.05$ was the significance threshold. Comparisons involving restoration were explicitly treated as comparisons between different terminal times, not paired changes. No cross-assay correlations were calculated because sample overlap could not be reconstructed.

Results

Regional biochemical differences at week 4

Model rats had lower regional SOD activity than concurrent sham rats: 0.564 ± 0.072 versus 1.122 ± 0.072 U/mg protein (Fig. 1A). The model-minus-sham difference was −0.558 U/mg protein, with a 95% Games–Howell confidence interval of −0.672 to −0.443 and adjusted $P < 0.001$. The model mean was approximately 49.7% lower than the sham mean.

The MDA-related TBA signal was higher in model than sham rats: 0.349 ± 0.034 versus 0.095 ± 0.017 nmol/mg protein (Fig. 1B). The model-minus-sham difference was 0.253 nmol/mg protein, with a 95% confidence interval of 0.209 to 0.298 and adjusted $P < 0.001$. Across the three terminal cohorts, Welch ANOVA yielded $F(2, 9.816) = 91.626$ for SOD and $F(2, 8.500) = 130.738$ for the MDA-related signal (both $P < 0.001$). Complete estimates and pairwise results are given in Supplementary Tables S1 and S2.

Exploratory molecular assay observations

Normalized *Nox1*-targeted assay estimates were 1.459 ± 0.156 in model samples and 1.000 ± 0.134 in sham samples, with Holm-adjusted P = 0.0037 (Fig. 2A). Corresponding *Nox4*-targeted estimates were 2.902 ± 0.558 and 1.000 ± 0.233, with adjusted P = 0.0001 (Fig. 2B). These differences describe the normalized outputs of the targeted assays under the stated efficiency and pairing assumptions. They do not confirm *Nox1*/*Nox4* transcript abundance because amplicon specificity remained unconfirmed.

Observations in the later restoration cohort

At week 8, four weeks after implant removal, restoration-cohort SOD activity was 1.035 ± 0.101 U/mg protein and the MDA-related signal was 0.213 ± 0.049 nmol/mg protein. Relative to week-4 model values, SOD was higher and the MDA-related signal lower (both adjusted P < 0.001). The MDA-related signal remained higher than the week-4 sham value (adjusted P = 0.0033); the SOD comparison with sham was nonsignificant (adjusted P = 0.2534).

Restoration-cohort *Nox1*/*Nox4*-targeted estimates were 1.180 ± 0.090 and 1.645 ± 0.227, respectively. Both were lower than model estimates (adjusted P = 0.0139 and 0.0037). Compared with sham, the *Nox4*-targeted estimate remained higher (adjusted P = 0.0099), whereas the *Nox1*-targeted difference was nonsignificant (adjusted P = 0.0618). Nonsignificant comparisons do not demonstrate equivalence or normalization. Because the cohorts differed in terminal time, these findings cannot separate implant removal from elapsed time, aging or second-surgery effects.

Discussion

The principal observation was a concurrent week-4 difference in regional medullary biochemistry: model rats had lower SOD activity and a higher MDA-related TBA signal than sham rats. Reporting both measurements adds a regional biochemical dimension to the established cervical model. The finding concerns a tissue block containing the RVLM and nearby medulla, rather than a defined neuronal population. It therefore provides a descriptive basis for more anatomically and chemically resolved experiments, without demonstrating that an RVLM-specific pathway mediated the cervical or blood-pressure phenotype.

The SOD result is relevant because reduced antioxidant-enzyme activity has been implicated experimentally in RVLM-associated hypertension. Chan and colleagues showed that restoring antioxidant enzymes in the RVLM lowered pressure in spontaneously hypertensive rats [5]. Our assay measured total SOD activity, however, and cannot distinguish SOD isoforms, enzyme abundance or the biochemical reasons for lower measured activity. The prior

intervention study supports biological plausibility; it does not make the current association causal or establish that SOD changes explain blood pressure in this model.

The MDA-related result addresses a different biochemical feature. Its direction alongside lower SOD activity is consistent with an altered regional redox profile, but the two assays do not validate each other. TBA-derived chromogens are not chemically specific to MDA [9], and protein normalization cannot remove all differences in tissue composition or reaction interference. Accordingly, the measured signal should not be presented as an absolute measure of lipid injury. The supplementary O12 index is even more restricted because tissue-matrix and buffer compatibility were not established. Common sample handling alone cannot exclude differential matrix effects.

The exploratory qPCR differences provide a question for subsequent molecular work rather than an answer about oxidase function. NOX1 and NOX4 differ in activation requirements and localization [8], so an RNA-level estimate cannot substitute for enzyme activity. Furthermore, NOX4 has shown protective effects in a vascular experimental setting [11], illustrating why a larger NOX-related signal cannot automatically be classified as pathogenic across tissues. In the present material, product specificity remains unconfirmed, amplification efficiency was assumed and a single unvalidated reference gene was used. These qualifications preclude a conclusion of established gene upregulation. No NOX enzyme-activity measurement or isoform-specific perturbation was available to connect the targeted assays to the biochemical differences.

Other regional processes remain possible. NADPH oxidase-dependent Ang II signaling can influence RVLM pressor responses in experimental studies [6], while mitochondrial impairment can also contribute to regional oxidant production [7]. The related Ang II/AT1-R report [2] supplies a reason to investigate these alternatives in this model, but shared program-level observations do not demonstrate mediation. The present data cannot identify a cervical afferent route, a particular ROS source, or a sequence from NOX activation through sympathetic output to elevated pressure. Mechanistic discrimination would require a validated region-specific intervention and physiological measurements obtained from the corresponding animals.

The restoration cohort is informative as a later endpoint. Its mean SOD activity was closer to the sham value, while the MDA-related signal remained elevated relative to sham. Describing these values preserves an observation that may help plan follow-up studies. It does not show that removing the implant reversed the biochemical phenotype. The model and sham cohorts ended at week 4, whereas restoration animals were studied at week 8. Differences therefore

include elapsed time, age, spontaneous evolution of the model, repeat anesthesia and surgery, in addition to implant removal. A continued-implant week-8 cohort and a time-matched second-surgery sham cohort are needed to distinguish these explanations.

Several measurement and design limitations affect the strength of inference. Fourteen of the 18 SOD inhibition values lay outside the manufacturer's suggested 40–60% pilot optimization interval, and corresponding dilution-validation records were not available. This recommendation is not a demonstrated hard validity boundary, but quantitative interpretation requires caution. The qPCR assays had one measurement per sample and lacked retained NTC/no-RT records and complete validation. Although investigators confirmed independent animal samples and target/reference pairing, the historical animal–sample ledger and cross-assay overlap were unavailable. The small, male-only samples and undocumented blinding further limit precision, generalizability and assessment of bias. These are limitations of the available evidence rather than reasons to remove inconvenient observations.

Animal welfare also affects interpretation. Nine of the 50 rats undergoing induction died, and no postoperative analgesics were given after induction or implant removal. A specific written justification or authorization for withholding analgesia was not retained. Perioperative observations and antibiotic treatment cannot establish adequate pain relief. Pain and stress are therefore potential influences on the measured regional endpoints, and ethical approval alone does not resolve the absence of a documented analgesia rationale. Any prospective extension of this work should use an institutionally reviewed analgesia plan and predefined monitoring and intervention criteria.

In conclusion, rats with atlantoaxial joint disorder showed lower regional medullary SOD activity and higher MDA-related TBA signals than concurrent sham rats at week 4. The later restoration cohort and targeted qPCR estimates extend the description but do not establish an implant-removal effect or a NOX-mediated mechanism. The useful next question is whether the regional biochemical difference persists with validated assays and time-matched controls, and whether a defined intervention links it to autonomic function.

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Conflict of Interest

The authors declare no competing interests.

Author contributions

Xiangying Yuan: Conceptualization, Investigation, Formal analysis, Writing – original draft. Bin Yang: Methodology, Investigation. Zongbao He: Investigation, Resources. Youkui Lv: Conceptualization, Supervision, Funding acquisition, Writing – review and editing. Zijian Wu: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review and editing.

Data availability

Processed biochemical values, paired target/reference Cq values, statistical results and the available original qPCR exports accompany this article. The Supplementary Information provides assay qualifications and the two plate-annotation explanations. Animal identity and cross-assay linkage records were not recoverable; reported biological replication and pairing rely on investigator confirmation. The original ethics approval document and approved protocol were not retained in the available archive. No cross-assay individual-animal linkage is asserted.

DECLARATION on the Use of AI

Codex (OpenAI; CLI 0.162.0-alpha.17.2) assisted with manuscript drafting, editing and data checking. The authors are responsible for reviewing and verifying the outputs and take full responsibility for the final manuscript.

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Figure 1

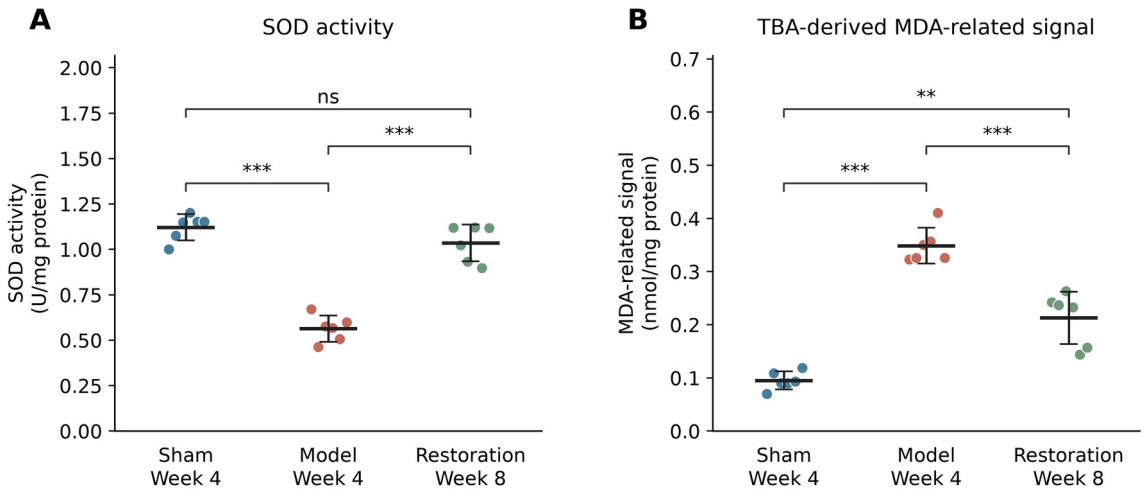


Fig. 1. Regional medullary SOD activity (A) and TBA-derived MDA-related signal (B). Points show the six independently processed samples per group; horizontal lines and error bars indicate mean ± sample SD. Sham and model cohorts were sampled at week 4; the independent restoration cohort was sampled at week 8 after implant removal at week 4. Brackets show Games–Howell comparisons following Welch ANOVA, with adjustment across the three comparisons within each endpoint: **P < 0.01, ***P < 0.001; ns, P ≥ 0.05. Comparisons involving restoration do not isolate an implant-removal effect. Tissue contained the RVLM and adjacent medulla.

Figure 2

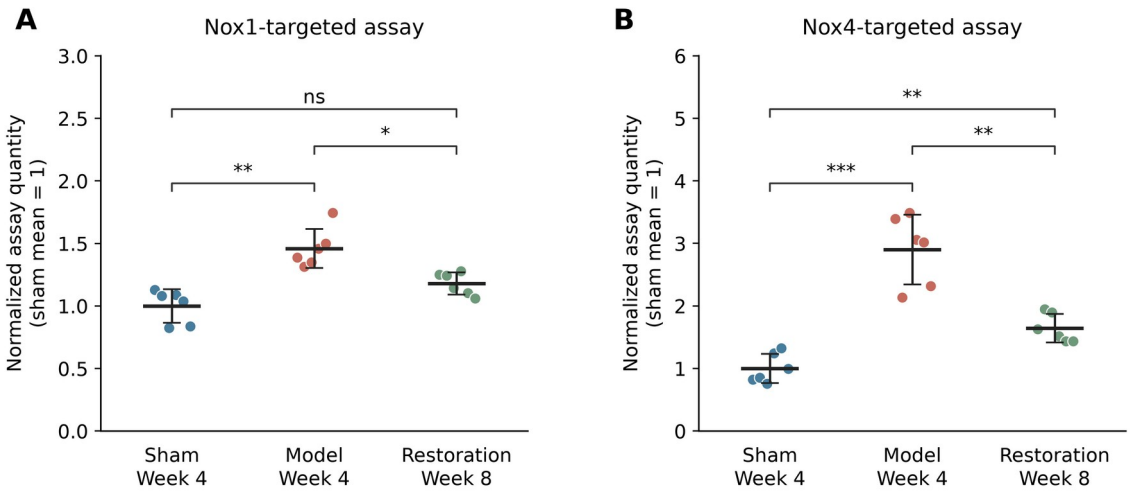


Fig. 2. Exploratory normalized *Nox1*-targeted (A) and *Nox4*-targeted (B) RT-qPCR assay estimates. Points show samples; horizontal lines and error bars indicate mean ± sample SD (n = 6/group). Quantities were normalized to the arithmetic mean quantity in sham samples; individual sham values were not set to 1. Statistics used ΔCq and two-sided Welch t tests, with Holm correction across the original 18-comparison family: *P < 0.05, **P < 0.01, ***P < 0.001; ns, P ≥ 0.05. Target specificity and amplification efficiency were not established. Sham/model sampling was at week 4 and restoration sampling at week 8.